

A Design Proposal of Silicon-on-Insulator Mach-Zehnder Interferometers

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Abstract— This report presents the design of a strip-waveguide Mach-Zehnder interferometer (MZI) for silicon-on-insulator (SOI) fabrication. The device is analyzed using a combination of analytical models and numerical simulations to quantify key performance metrics, including phase response, interference behavior, and transmission characteristics. The results are used to justify the chosen design parameters and to provide insight into the trade-offs that govern SOI MZI performance.

Index Terms— Silicon Photonics, Mach-Zehnder interferometer

I. INTRODUCTION

Silicon-on-insulator (SOI) photonics has become a leading integrated photonics platform because it enables high-index-contrast waveguides, compact passive devices, and compatibility with CMOS fabrication [1] [2]. Among the fundamental building blocks in SOI is the Mach-Zehnder interferometer (MZI), which converts optical phase differences into measurable intensity variations through interference. MZIs are widely used for optical modulation, switching, sensing, filtering, and characterization of waveguide phase and loss [3][5].

In an SOI MZI, light from an input waveguide is split into two arms, accumulates a relative phase delay due to differences in effective index and/or path length, and is then recombined at a second coupler. The resulting transmission depends on the phase difference $\Delta\phi$, which is wavelength-dependent through the propagation constant $\beta(\lambda) = 2\pi n_{eff}(\lambda)/\lambda$. Accurate modeling therefore requires capturing both the waveguide dispersion (variation of n_{eff} with wavelength) and practical non-idealities such as propagation loss and imperfect splitting ratios. For imbalanced MZIs, these effects produce spectral fringes with a free spectral range (FSR) that is governed by the arm length difference and the waveguide group index.

This report presents the modeling of an SOI Mach-Zehnder interferometer using a compact waveguide model and an analytical interferometer transfer function. The simulation predicts the optical transmission spectrum around 1550 nm, and key performance metrics such as spectral variation across a wavelength sweep, and sensitivity to design parameters (arm length difference). The results provide both physical insight into MZI behavior on SOI and practical guidance for designing broadband splitters and interferometric devices in silicon photonic circuits.

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II. THEORY

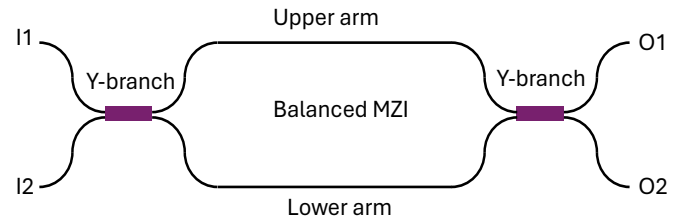


Fig. 1 Functional layout of a balanced MZI.

We consider the balanced Mach-Zehnder interferometer shown in Fig. 1, implemented with two ideal 3-dB Y-branches acting as a splitter and a combiner. Let the complex input fields at the two input ports be E_{I1} and E_{I2} . An ideal 3-dB Y-branch splitter launches equal-amplitude fields into the two arms; using a symmetric representation, the fields at the entrance of the upper and lower arms can be written as:

$$E_{u,l}(0) = \frac{E_{I1} + E_{I2}}{\sqrt{2}} \quad (1)$$

The fields then propagate through the upper and lower arms of lengths, L_u and L_l , accumulating phase according to the waveguide propagation constant:

$$\beta(\lambda) = \frac{2 \cdot \pi \cdot n_{eff}(\lambda)}{\lambda} \quad (2)$$

Neglecting any additional phase-shifting element, the fields at the input of the output Y-branch are:

$$E_u(L_u) = E_u(0) \cdot e^{-i\beta_u L_u - \frac{a_u}{2} L_u} \quad (3)$$

$$E_l(L_l) = E_l(0) \cdot e^{-i\beta_l L_l - \frac{a_l}{2} L_l} \quad (4)$$

, where a_u and a_l the propagation loss experienced into the two arms. Assuming that only input 1 is active ($E_{O2} = 0$) and neglect any propagation loss ($a_u = a_l = 0$), at the output port of the second Y-branch, the fields are:

$$E_{O1,2} = \frac{E_u(L_u) + E_l(L_l)}{\sqrt{2}} \quad (5)$$

$$E_{O1,2} = \frac{1}{2} \cdot E_{I1} (e^{-i\beta_u L_u} + e^{-i\beta_l L_l}) \quad (6)$$

As the light intensity is proportional to the squared power of the field amplitude, the light intensity at the output of the MZI can

be written as:

$$I_{01,2} = \frac{1}{4} \cdot |E_{I1}(e^{-i\beta_u L_u} + e^{-i\beta_l L_l})|^2 \quad (7)$$

Given that both interferometer arms use the same waveguide geometry and no additional optical phase shifter is included, the propagation constant are equal:

$$\beta_u = \beta_l = \frac{2 \cdot \pi \cdot n_{eff,u}(\lambda)}{\lambda} = \frac{2 \cdot \pi \cdot n_{eff,l}(\lambda)}{\lambda} = \beta \quad (8)$$

In this way, eq. (7) is simplified to:

$$I_{01,2} = \frac{1}{2} \cdot I_{I1} \cdot (1 + \cos(\beta \Delta L)) \quad (9)$$

, where $I_{I1} = E_{I1}^2$ and $\Delta L = L_u - L_l$.

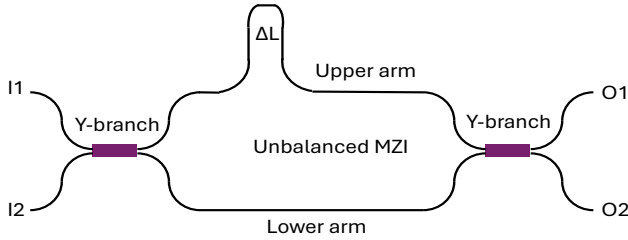


Fig. 2 Functional layout of an unbalanced MZI.

Fig. 2 shows the functional layout of an MZI in which the two arms have different optical path lengths ($L_u \neq L_l$), so the relative phase between the two propagated fields becomes strongly wavelength-dependent. When the fields recombine at the output coupler, this varying phase produces alternating constructive and destructive interference, resulting in a sinusoidal transmission spectrum with periodic maxima and minima; the spacing of these fringes is the free spectral range (FSR):

$$FSR(\lambda) = \frac{\lambda^2}{\Delta L \cdot n_g(\lambda)} \quad (10)$$

, where n_g is the group index defined as:

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda \frac{dn_{eff}(\lambda)}{d\lambda} \quad (11)$$

III. MODELLING AND SIMULATION

For the simulation studies, the standard strip-waveguide geometries was considered. Table 1 summarizes the silicon core dimension (width and height) together with the surrounding silicon-dioxide cladding thickness. For completeness and reproducibility, the key finite-difference eigenmode (FDE) solver settings used in the analysis are also listed.

Table 1 Simulated waveguide geometries.

Obj.	Geometry Param.	Design 1
Si	Width span (nm)	500
	Height span (nm)	220
SiO2	Width span (nm)	4
	Height span (nm)	4
FDE solver	Width span (nm)	2.5
	Height span (nm)	2

Mesh cells x	100
Mesh cells y	100

A. Strip waveguide with 500 nm width

The following section presents the characteristics of waveguide geometry.

Table 2 Effective, group indices and loss for TE1 and TM1

$\lambda = 1500$ nm	Neff	Group index
TE1	2.506654	4.197804
TM1	1.837414	4.021323

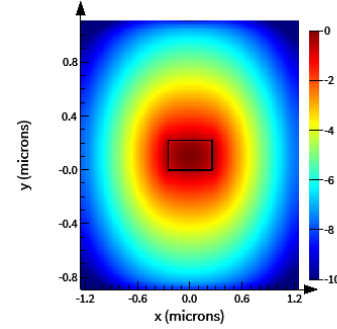


Fig. 3 Field intensity of TE1.

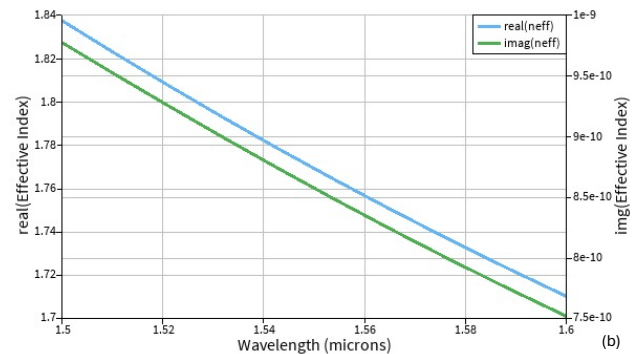
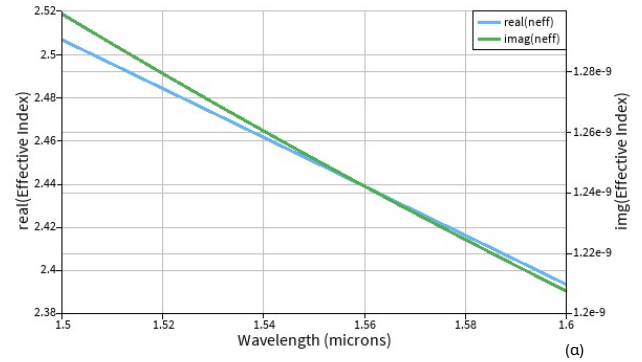


Fig. 4 Effective indices over wavelength for TE1 and TM1.

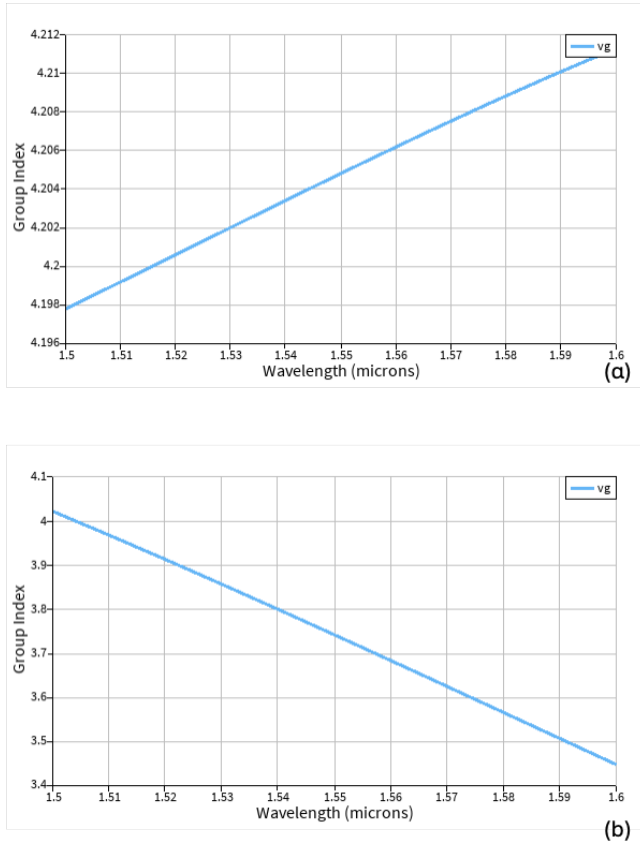


Fig. 5 Group indices over wavelength for TE1 and TM1.

The compact models of this waveguide geometry for TE1 and TM1 modes are:

$$n_{eff,TE1}(\lambda) = 2.45017 - 1.13197 \cdot (\lambda - \lambda_0) - 0.0441148 \cdot (\lambda - \lambda_0)^2 \quad (12)$$

$$n_{eff,TM1}(\lambda) = 1.76917 - 1.27212 \cdot (\lambda - \lambda_0) + 1.86092 \cdot (\lambda - \lambda_0)^2 \quad (13)$$

B. Mach-Zehnder Interferometer designs

For the design of the unbalanced MZIs, we select to work with the TE mode of the three waveguide geometries. Table 3 presents the simulated FSRs for length asymmetries 0 μm , 25 μm , 50 μm , 75 μm and 100 μm .

Table 3 Simulated FSR of unbalanced MZI based for $\lambda = 1500$ nm and $\lambda = 1556.6$ nm for the selected waveguide geometry.

ΔL (μm)	FSR (nm) $\lambda = 1500$ nm
0	No fringes
30	17.87
50	10.72
70	7.66
90	5.96
110	4.87
130	4.12

Fig 6 presents the simulation setup developed for the purposes of the current report emulating a MZI.

Fig. 7 and Fig 8 depict the transfer function of the grating couplers and Y-branched, respectively. As expected, both devices are wavelength-dependent show a strong attenuation as we move away from the central wavelength of 1550 nm.

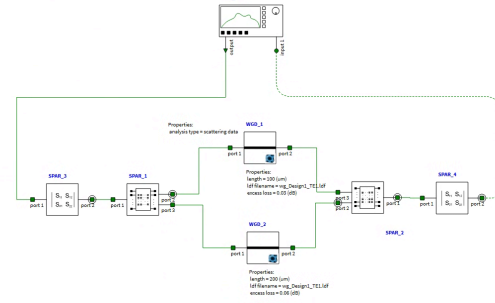


Fig. 6 Simulation setup in Interconnect.

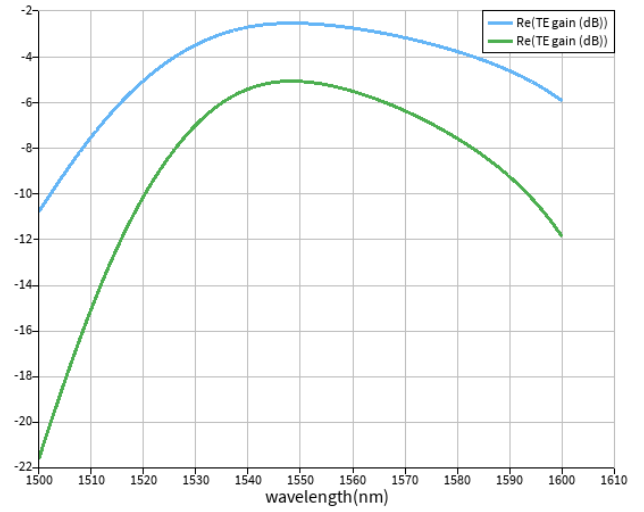


Fig. 7 Transfer function of a single (blue) and dual (green) GCs.

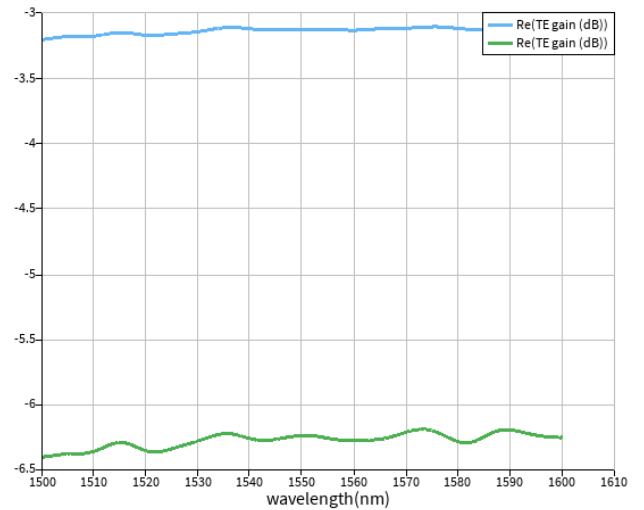


Fig. 8 Transfer function of a single (blue) and dual (green) Y-branches.

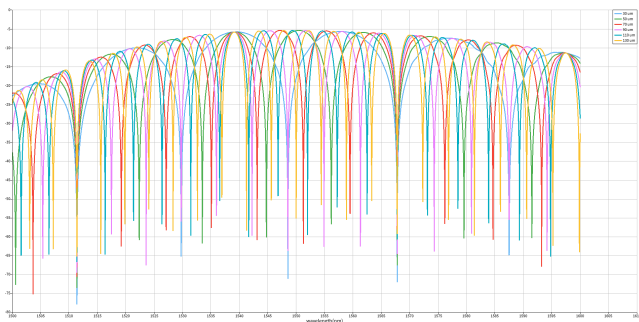


Fig. 9 Transfer function of unbalanced MZIs with 25 μm , 50 μm , 75 μm and 100 μm length asymmetry using Design-1 waveguide geometry.

Finally, Fig 9 presents the transfer function of the simulated and designed unbalanced MZIs from 1500 nm to 1600 nm.

V. MASK LAYOUT

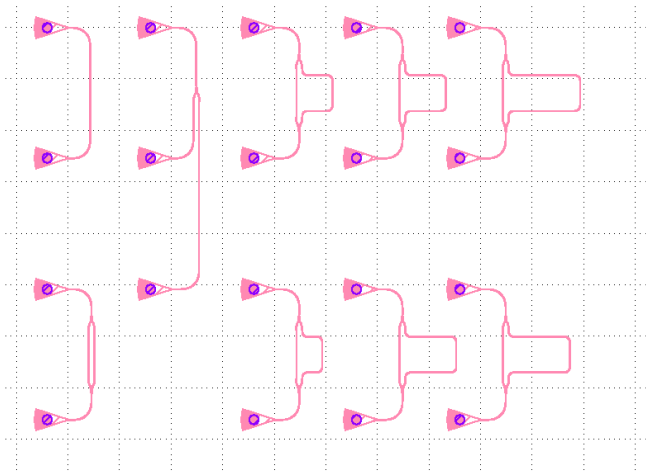


Fig. 10 Mask layout.

Fig. 10 Mask layout.illustrates the submitted mask layout. The design consists of a cascade of unbalanced MZIs implemented on a silicon photonics platform. Optical input and output are achieved by utilizing grating couplers (GCs) positioned at the periphery of the layout, enabling efficient fiber-to-chip coupling. Within each interferometric stage, compact Y-branch splitters and combiners are used to divide and recombine the optical signals. The waveguide routing is carefully folded to minimize the overall footprint while maintaining low-loss connections between components. Each MZI comprises two arms of unequal length, introducing a controlled optical path difference (ΔL) that governs the interference behavior.

In addition to the cascaded interferometric structure, dedicated test structures were included to enable independent characterization of key building blocks. To evaluate the performance of the GCs, a reference circuit consisting of a pair of GCs connected back-to-back was implemented. This configuration allows direct extraction of coupling efficiency and spectral response by isolating GC losses from other components. Similarly, a standalone dual Y-branch structure,

terminated with grating couplers at all ports, was designed to characterize splitting ratio, excess loss, and wavelength dependence. These auxiliary test structures are essential for decoupling component-level performance from the overall MZI response, thereby improving the accuracy of model validation and enabling more reliable interpretation of the interferometric measurements. This layout therefore provides a compact and effective test structure for experimentally validating interferometric behavior, extracting the waveguide group index, and assessing the accuracy of compact models used in the design of wavelength-dependent silicon photonic circuits.

VI. CORNER ANALYSIS

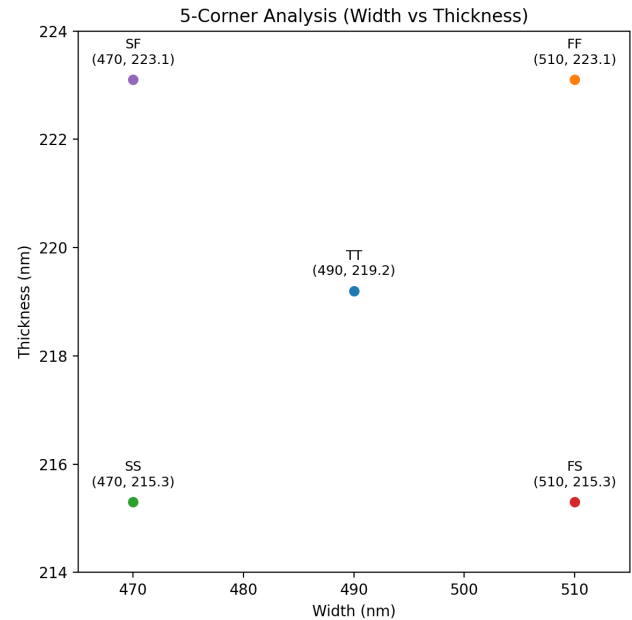


Fig. 11 5-point corner analysis grid.

Corner analysis is a systematic approach used to evaluate how fabrication variations affect the performance of a photonic device. In silicon photonics, the physical dimensions of the waveguide, primarily its width and thickness, cannot be perfectly controlled during manufacturing. Small deviations arise from lithography, etching, and material non-uniformities. Because the optical properties of the waveguide, such as the effective index and group index, depend strongly on these dimensions, even nanometer-scale variations can lead to measurable changes in device behavior. Rather than treating these variations as random, corner analysis selects a small number of representative cases that capture the extremes and the nominal condition of the fabrication process. The goal is to verify that the device operates correctly under all realistic conditions without requiring a full statistical simulation.

In this context, the fabrication space is defined by two independent variables: the waveguide width, which varies between 470 nm and 510 nm, and the silicon thickness, which

Table 4 Corner analysis of effective and group indices at 1550 nm.

Corner Type	Width (nm)	Thickness (nm)	n_{eff}	n_g
Nominal (TT)	500	220	2.4404	4.17694

Min/Min (SS)	470	215.3	2.3703	4.22857
Max/Max (FF)	510	223.1	2.46605	4.16548
Min W/ Max T (SF)	470	223.1	2.39866	4.23711
Max W/Min T (FS)	510	215.3	2.43796	4.15844

varies between 215.3 nm and 223.1 nm. The midpoint of each range represents the nominal fabrication condition, corresponding to a width of 500 nm and a thickness of 220 nm. These limits define a two-dimensional parameter space in which every fabricated device can be located. One can visualize this space as a grid, where the horizontal axis corresponds to width and the vertical axis corresponds to thickness. Each axis contains three key values: a minimum, a nominal, and a maximum. In principle, combining all these values produces nine possible cases, forming a 3×3 grid.

However, in practice, it is not necessary to simulate all nine combinations. A reduced set of five cases is typically sufficient to capture the most important physical effects. These are known as the five process corners. The central point of the grid corresponds to the nominal condition, where both width and thickness take their midpoint values. Two of the corners lie along the diagonal of the grid: one where both parameters are at their maximum values and one where both are at their minimum values. These represent situations where the fabrication variations are correlated, meaning both dimensions shift in the same direction. The remaining two corners lie on the opposite diagonal, where one parameter is at its maximum while the other is at its minimum. These represent uncorrelated or asymmetric variations.

This grid representation is important because it reflects how physical changes in geometry translate into optical behavior. When both width and thickness increase together, the optical mode becomes more confined, typically leading to an increase in effective index. When both decrease, confinement is reduced and the effective index decreases. These cases mainly shift the overall optical response of the device. In contrast, when one parameter increases and the other decreases, the effect is not simply a uniform shift. Instead, it introduces a form of imbalance, because width and thickness do not influence the optical mode in exactly the same way. As a result, these asymmetric cases often produce the most critical performance degradation.

This is particularly important in devices that rely on phase matching or interference, such as Mach–Zehnder interferometers or ring resonators. In such devices, performance depends on precise control of the optical phase. Variations in effective index directly translate into phase errors, and these errors can accumulate over the length of the device. The asymmetric corners are often responsible for the worst-case phase mismatch, even if their dimensional deviations are not the largest in absolute terms. For this reason, corner analysis is not just a geometric exercise but a physically meaningful way to probe device sensitivity.

Using the extracted effective index and group index values at each corner, the corner analysis can now be extended from a purely geometric interpretation to a quantitative assessment of optical sensitivity. The results show a clear dependence of both

n_{eff} and n_g on the combined variations of width and thickness, and more importantly, they reveal how different types of variation—correlated versus asymmetric—impact the optical response differently.

Starting from the nominal condition, the effective index is 2.4404. When both dimensions decrease simultaneously (minimum width and minimum thickness), the effective index drops significantly to 2.3703. This represents the largest negative deviation from nominal, corresponding to a reduction of approximately 0.070. Physically, this is expected, as both reduced width and reduced thickness weaken optical confinement, allowing the mode to spread more into the cladding. Conversely, when both dimensions increase, the effective index rises to 2.46605, an increase of about 0.026 from nominal. This asymmetry in variation—where the decrease is larger than the increase—indicates a nonlinear dependence of effective index on geometry, which is typical in high-index-contrast waveguides.

The asymmetric corners provide additional insight. When the width is reduced but the thickness is increased, the effective index becomes 2.39866, which is still significantly lower than nominal. This suggests that width variations have a stronger influence on the effective index than thickness variations in this geometry. In contrast, when the width is increased and the thickness is reduced, the effective index is 2.43796, which is very close to the nominal value. This further confirms that the sensitivity to width dominates over thickness, and that an increase in width can partially compensate for a decrease in thickness, but not vice versa.

A similar but not identical trend is observed for the group index. The nominal group index is 4.17694. At the minimum/minimum corner, the group index increases to 4.22857, indicating stronger dispersion when the mode is less confined. At the maximum/maximum corner, the group index decreases slightly to 4.16548. The asymmetric cases again show that the largest deviation occurs when the width is reduced, particularly in the minimum width / maximum thickness case, where the group index reaches 4.23711, the highest value among all corners. This reinforces the conclusion that width variations dominate the optical behavior, especially in terms of dispersion.

From a device perspective, these variations have direct implications. Changes in effective index translate into phase errors, which scale with device length. The observed range of effective index variation, spanning approximately 0.096 between the extreme corners, can lead to substantial phase deviations in interferometric devices. In particular, the minimum/minimum corner represents the worst-case scenario for phase delay reduction, while the maximum/maximum corner produces a smaller but still non-negligible increase. The asymmetric corners, although not always producing the largest absolute index shift, are critical because they can introduce imbalance between different sections of a circuit.

The variation in group index is equally important for wavelength-dependent behavior. Since the group index determines the propagation delay and spectral response,

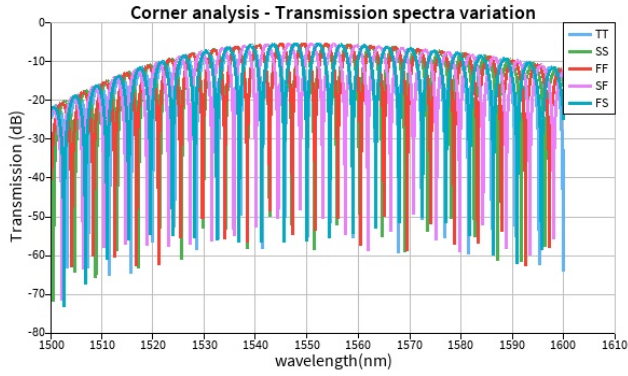


Fig. 12 Corner analysis of aMZI with $AL = 130$ nm - TF variation.

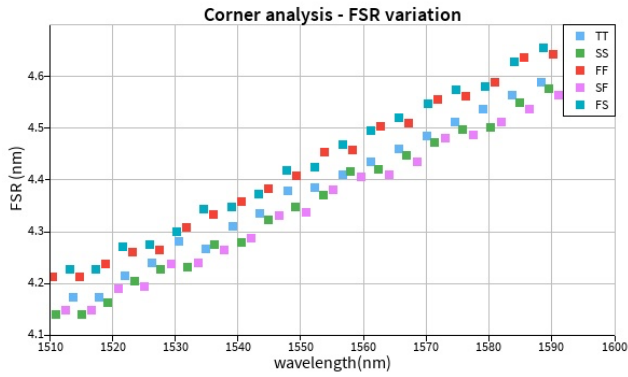


Fig. 13 Corner analysis of aMZI with $AL = 130$ nm - FSR variation.

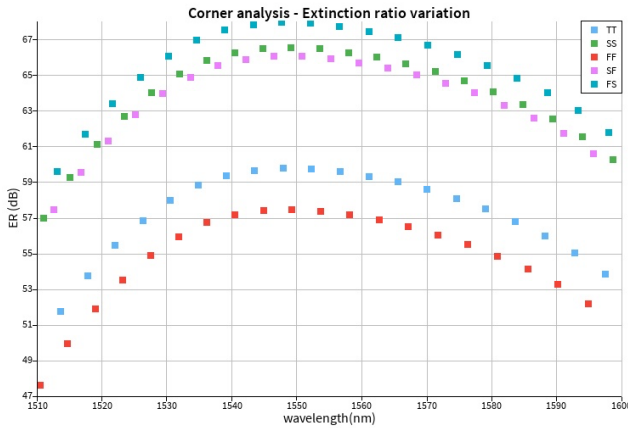


Fig. 14 Corner analysis of aMZI with $AL = 130$ nm - ER variation.

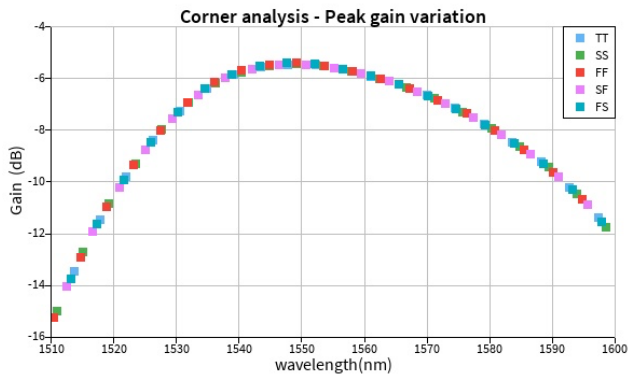


Fig. 15 Corner analysis of aMZI with $AL = 130$ nm - peak gain variation.

differences across corners will lead to shifts in resonance wavelength and free spectral range. The fact that the group index varies by approximately 0.08 across corners indicates that dispersion-related effects cannot be neglected, especially in broadband or resonant devices.

Overall, this analysis shows that the device is significantly more sensitive to reductions in width than to increases, and that thickness variations play a secondary but still noticeable role. The most critical corners are those involving minimum width, particularly when combined with thickness variations, as they produce the largest deviations in both effective and group index. This suggests that fabrication control of width is the dominant factor in ensuring device stability.

The spectral results obtained from the aMZI simulations provide a direct validation of the conclusions drawn from the effective and group index analysis, and they illustrate how geometric variations translate into observable optical behavior. When these plots are interpreted together with the previously extracted n_{eff} and n_g , a consistent and physically meaningful picture emerges.

The transmission spectra clearly show that all corners preserve the same overall interference pattern, but the position of the fringes is shifted along the wavelength axis. This behavior is a direct consequence of variations in the effective index. Since the phase accumulated in each arm depends on n_{eff} , any deviation from the nominal value results in a global shift of the interference condition. Corners with higher effective index shift the spectrum toward longer wavelengths, while those with lower effective index shift it toward shorter wavelengths. This is consistent with the earlier observation that the maximum/maximum corner produces the highest effective index and therefore the largest red shift, whereas the minimum/minimum corner produces the lowest effective index and a corresponding blue shift. The key implication is that fabrication variations do not alter the fundamental shape of the spectral response, but rather translate it along the wavelength axis.

In addition to this spectral shift, the spacing between adjacent fringes is also affected by process variations, as shown in the FSR plot. The FSR increases with wavelength for all corners, reflecting the dispersive nature of the waveguide. However, there is a clear separation between corners, which is directly linked to differences in group index. Since the FSR is inversely proportional to n_g , corners with higher group index produce smaller spectral spacing, while those with lower group index produce larger spacing. This behavior matches the earlier numerical analysis, where the minimum-width configurations exhibited the highest group index and therefore the smallest FSR. The variation is relatively small but systematic, indicating that fabrication introduces a predictable spread in spectral periodicity rather than random distortion.

The extinction ratio results further highlight the impact of fabrication-induced phase imbalance. Ideally, an interferometer achieves maximum extinction when the two arms recombine with equal amplitude and a precise π phase difference at destructive interference points. The observed variation in extinction ratio across corners indicates that this condition is not equally satisfied in all cases. While the general trend of the extinction ratio remains similar across wavelengths, the peak values differ noticeably between corners. This confirms that

geometric variations introduce phase errors that prevent perfect cancellation. Importantly, the degradation is not uniform, suggesting that some corners are closer to ideal phase matching than others. This reinforces the earlier conclusion that asymmetric variations, particularly those involving reduced width, can introduce critical imbalance even when the absolute index change is not maximal.

In contrast, the peak gain results show minimal variation across all corners. The envelope of the spectral response remains nearly unchanged, indicating that propagation loss and overall transmission efficiency are not significantly affected by the considered geometric variations. This observation is important because it isolates the dominant effect of fabrication variability: the device is primarily phase-sensitive rather than loss-sensitive. In other words, the main impact of process variations is on interference conditions rather than on attenuation.

Taken together, these plots confirm that the behavior of the aMZI under fabrication variations can be understood as the combined effect of two mechanisms. Variations in effective index produce a global shift of the spectral response, while variations in group index modify the spacing between interference fringes. These two effects occur simultaneously and determine the final spectral characteristics of the device. The extinction ratio adds a third dimension, revealing how deviations from ideal phase matching degrade interference contrast.

Overall, the spectral analysis reinforces the conclusion that width variations are the dominant factor influencing device performance. Corners involving reduced width consistently produce the largest deviations in both spectral position and fringe spacing, as well as noticeable changes in extinction ratio. Thickness variations play a secondary role, modifying the response but not dominating it. From a design perspective, this means that controlling waveguide width is critical for ensuring spectral stability and maintaining high interference quality.

VII. FABRICATION PROCESS

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices

were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapor deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

VIII. EXPERIMENTAL PERFORMANCE EVALUATION

To characterize the devices, a custom-built automated test setup [6] [10] with automated control software written in Python was used [7]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [8]. A 90° rotation was used to inject light into the TM grating couplers [8]. A polarization maintaining fibre array was used to couple light in/out of the chip [9].

Prior to analyzing the interferometric structures, the individual building blocks were experimentally characterized to isolate their contributions to the overall circuit response. Fig. 16 shows the transmission spectrum of two cascaded GCs. The response exhibits a typical broadband Gaussian-like behavior, with a peak transmission occurring at a wavelength of 1.5499 μm . From this measurement, the minimum insertion loss per grating coupler is extracted to be approximately -7.78 dB. The relatively high loss is consistent with standard surface grating couplers and reflects coupling inefficiencies as well as mode mismatch between the optical fiber and the on-chip waveguide. The spectral shape also confirms the wavelength-selective nature of the GCs, which must be considered when interpreting the overall system response.

Fig. 17 further presents the transmission characteristics of a standalone Y-branch splitter. The measured transmission is approximately -3.1 to -3.6 dB across the wavelength range of interest, which is close to the ideal -3 dB splitting ratio expected for a 50:50 power divider. The slight excess loss and wavelength dependence can be attributed to fabrication imperfections, mode mismatch at the junction, and bending losses. These results confirm that the Y-branch operates as an approximately balanced splitter, suitable for use in interferometric configurations.

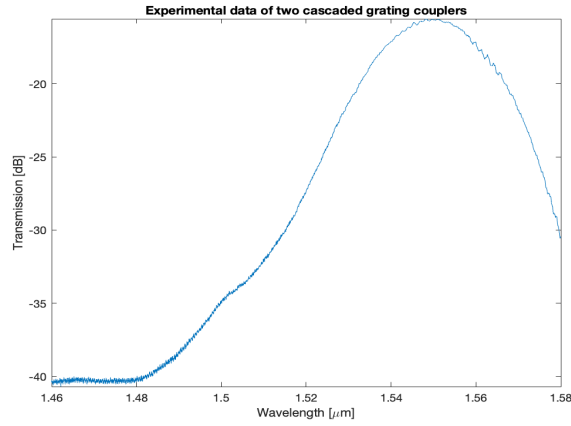


Fig. 16 Transfer function of two cascaded connected grating couplers.

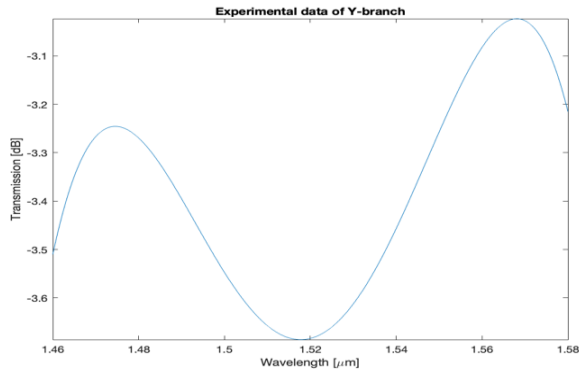


Fig. 17 Transfer function of Y-branch.

From the different variations of the unbalanced MZI was selected the MZI with 130 μm length asymmetry to be analyzed. Fig. 18 presents the raw experimental data from the Scylla dataset. The experimental transmission spectrum of the fabricated MZI was first processed baseline correction to remove slow variations introduced by the measurement setup and grating couplers. The corrected spectrum, shown in Fig. 19, exhibits a clear periodic interference pattern over the wavelength range 1.53–1.57 μm , with extinction ratios exceeding 25 dB. The high visibility of the fringes confirms good splitting balance and low excess loss in the interferometer. The spectral periodicity corresponds to the FSR, which is directly related to the optical path length difference and the group index of the waveguide.

To extract the spectral features quantitatively, the MATLAB findpeaks algorithm was applied to the inverted and smoothed transmission signal in order to identify the minima of the interference pattern (Fig. 20). The spacing between consecutive minima was used to compute the FSR as a function of wavelength. As shown in Fig. 20, the FSR exhibits a slight increase with wavelength, ranging from approximately 4.32 nm to 4.50 nm. This trend reflects the dispersive nature of the waveguide, since the FSR is inversely proportional to the group index. Using the extracted FSR values and the known path length difference ($\Delta L = 130 \mu\text{m}$), the group index was calculated, yielding values around $n_g \approx 4.18$ – 4.20 with a weak decreasing trend versus wavelength, as shown in Fig. 22.

These extracted parameters were then used as initial conditions for nonlinear curve fitting of the full MZI transfer function. The

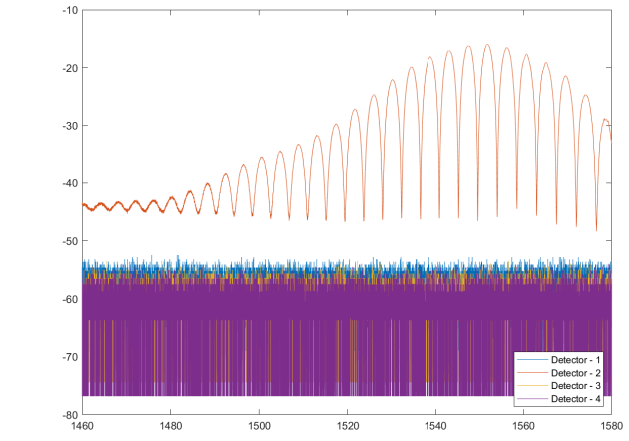


Fig. 18 Experimental results of unbalanced MZI with $\Delta L = 130 \mu\text{m}$

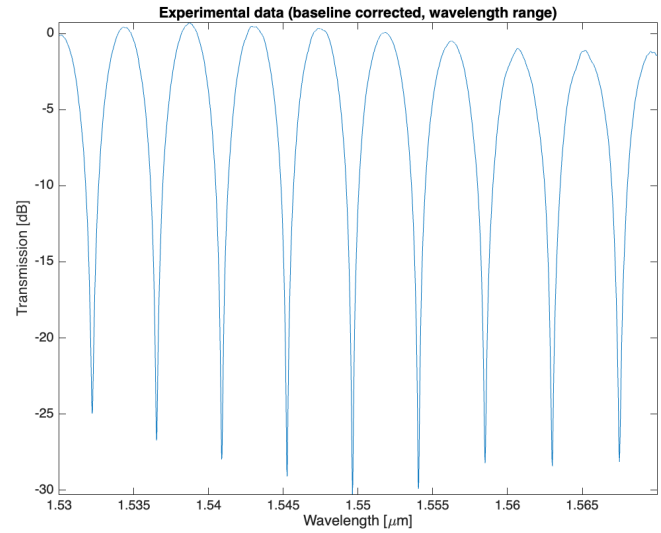


Fig. 19 Experimental results of unbalanced MZI with $\Delta L = 130 \mu\text{m}$ after baseline correction.

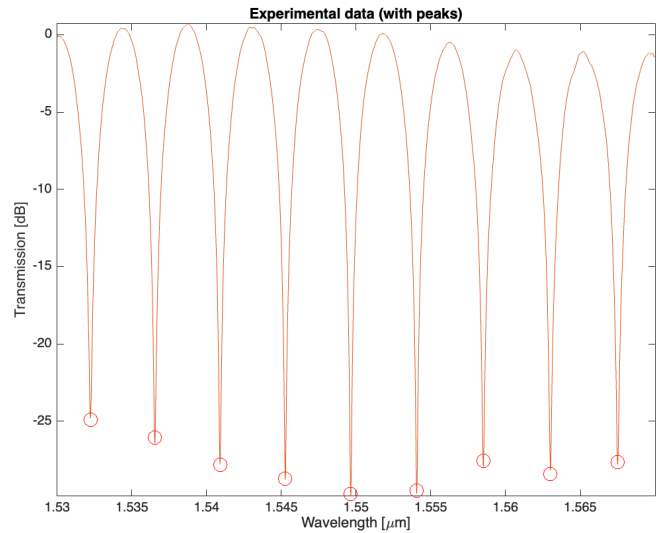
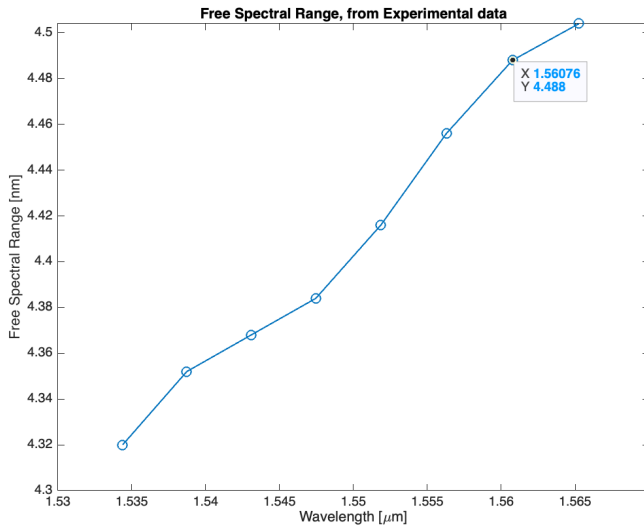
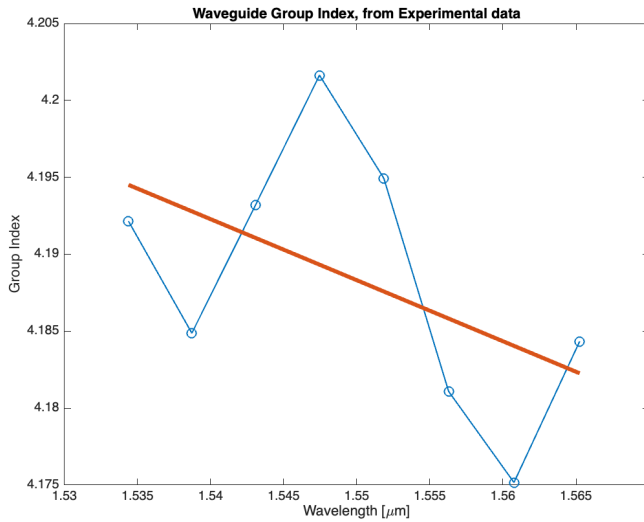
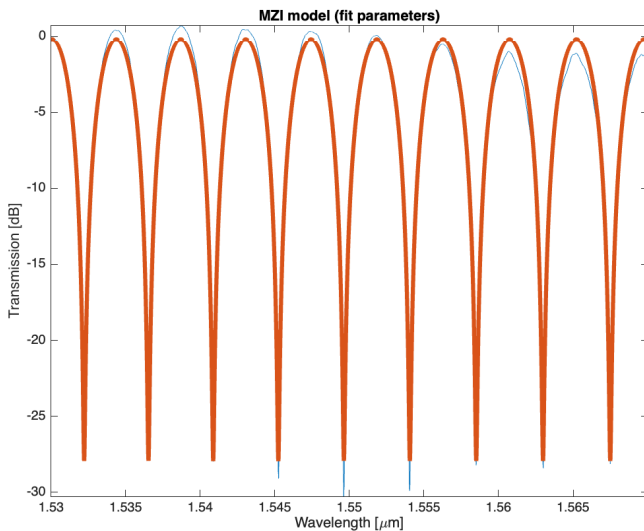


Fig. 20 Experimental results of unbalanced MZI with $\Delta L = 130 \mu\text{m}$ with selected minimum dips.


 Fig. 21 FSR of unbalanced MZI with $\Delta L = 130 \mu\text{m}$

 Fig. 22 Group index of MZI with $\Delta L = 130 \mu\text{m}$.

 Fig. 23 Fitting of MZI $\Delta L = 130 \mu\text{m}$.

included in the model. The fitted response (Fig. 23) shows excellent agreement with the experimental data, accurately reproducing both the fringe positions and extinction depths, indicating a high-quality fit ($R^2 > 0.8$).

The experimentally extracted group index values and spectral behavior are consistent with the trends predicted in the earlier corner analysis. In particular, the measured group index around $n_g \approx 4.18\text{--}4.20$ falls within the expected range for the nominal waveguide geometry, confirming that the fabricated devices are close to the intended design point. The slight wavelength dependence and gradual decrease of n_g are also in agreement with the dispersion characteristics obtained from the compact model. Furthermore, the observed FSR values align well with the predicted sensitivity of the interferometer to geometric variations, as previously discussed. Small deviations between experiment and model can be attributed to fabrication-induced variations in waveguide width and thickness, which were shown in the corner analysis to have a non-negligible impact on both effective and group indices. Overall, the agreement between simulation, corner analysis, and experimental results validates the robustness of the design and highlights the importance of accounting for process variations in silicon photonic circuits.

IX. CONCLUSION

This work presented the design, modeling, fabrication, and experimental characterization of a strip-waveguide MZI on a silicon-on-insulator platform. A comprehensive approach combining analytical modeling, numerical simulation, and experimental validation was employed to investigate the interferometric behavior and extract key waveguide parameters. The theoretical framework successfully described the relationship between phase accumulation, effective index, and spectral response, while simulations provided insight into the dependence of FSR on path length imbalance. The fabricated devices were experimentally characterized through a systematic methodology, beginning with the independent evaluation of fundamental building blocks, including grating couplers and Y-branch splitters. The measured performance of these components confirmed expected behavior, enabling accurate interpretation of the MZI response. The interferometric measurements demonstrated high extinction ratios and well-defined spectral fringes, indicating good device performance and low imbalance. Using peak detection and curve fitting techniques, the group index was extracted with high accuracy, yielding values in the range of $n_g \approx 4.18\text{--}4.20$, in close agreement with simulation and compact model predictions. Furthermore, the experimental results were shown to be consistent with the trends identified in the corner analysis, confirming the sensitivity of device performance to fabrication-induced variations. In particular, the dominant influence of waveguide width on both effective and group indices was validated experimentally. The combination of modeling, corner analysis, and measurement demonstrates that the designed MZI is robust against typical process variations while maintaining predictable spectral behavior.

effective index was modeled by a second-order Taylor expansion, while propagation loss and phase offset were also

ACKNOWLEDGMENTS

We acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Mateo Branion-Calles performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

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